Script Block and Functions

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Script Block

Collection of statements within curly braces
Example
HERE we get the output as it is - this is because no service runs by default as it is
If we want to print the output we need to use invoke command or put the complete command in the variable and then call the variable
{
Write-host : "this is a script-block" -foregroundcolor yellow
Get-service -name winrm
}

```
PS C:\Users\lenovo> {
>> Write-host : "this is a script-block" -foregroundcolor yellow
>> Get-service -name winrm
>> }
>>
>>

Write-host : "this is a script-block" -foregroundcolor yellow
Get-service -name winrm
```

Using Invoke Command

Invoke-command -ScriptBlock {
write-host "this is a script block" - foregroundcolor yellow
Get-service -name winrm
}

```
PS C:\Users\lenovo> Invoke-command  -ScriptBlock {
>>          write-host "this is a script block" - foregroundcolor yellow
>>          Get-service   -name winrm
>> }
>>
this is a script block - foregroundcolor yellow

Status   Name               DisplayName
------   ----               -----------
Stopped  winrm              Windows Remote Management (WS-Manag...
```

Using Variable Command

$sub = {
Write-host : "this is a script-block" -foregroundcolor yellow
Get-service -name winrm
}

```
PS C:\Users\lenovo> $sub = {
>> Write-host : "this is a script-block" -foregroundcolor yellow
>> Get-service -name bits
>> }
>>
PS C:\Users\lenovo> $sub

Write-host : "this is a script-block" -foregroundcolor yellow
Get-service -name bits

PS C:\Users\lenovo> $sub.invoke()
: this is a script-block

Status   Name               DisplayName
------   ----               -----------
Stopped  bits               Background Intelligent Transfer Ser...
```

Here if we Want to run the command present inside the {} then we need to use invoke command along with created variable.the command used previously
Get-service | where-object {$_.status -eq "running"} | select-object last 10

Functions

A**PowerShell function** is a named, reusable block of code that performs a specific task, similar in behavior to a built-in cmdlet . Functions help organize scripts, avoid code duplication, and make code easier to read, maintain, and troubleshoot.

It even reduces the code size.
Function show-text{
write-host "this is simple function"
}
Or
show-text{
write-host "this is simple function"
}

Make sure the name of the function(show-text) should be in Verb - Noun Format.
Once we enter , it is saved in power shell memory.

```
S C:\Users\lenovo> function show-text{
> write-host "this is simple function"
> }
S C:\Users\lenovo> show-text
his is simple function
S C:\Users\lenovo>
```

Once we change anything in the existing function it will not change in memory . To save it, we need to run function as shown type -show-text to run the function

```
S C:\Users\lenovo> function show-text{
> write-host "this is simple function"
> get-service -name WinRM
> }
S C:\Users\lenovo> show-text
his is simple function

tatus   Name               DisplayName
-----   ----               -----------
topped  WinRM              Windows Remote Management (WS-Manag...
```

```
PS C:\Users\lenovo> function show-text{
>> write-host "this is simple function" -foregroundcolor red
>> get-service -name WinRM
>> }
PS C:\Users\lenovo> show-text
this is simple function

Status   Name               DisplayName
------   ----               -----------
Stopped  WinRM              Windows Remote Management (WS-Manag...
```

TO PASS PARAMETER IN THE FUNCTION

There are 2 methods to pass parameter in the function

1st method (not recommended)

Function show-text($message, $servicename){
write-host "$message" -foregroundcolor red
get-service -name $servicename
}

To run

Show-text -message "coming from param1" -servicename "Bits"

```
PS C:\Users\lenovo> function show-text($message, $servicename){
>> write-host "$message" -foregroundcolor red
>> get-service -name $servicename
>> }
PS C:\Users\lenovo> show-text -message "coming from param1" -sercvicename "Bits"
coming from param1

Status   Name               DisplayName
------   ----               -----------
Stopped  AarSvc_4bf2e02     Agent Activation Runtime_4bf2e02
Running  AdobeARMservice    Adobe Acrobat Update Service
Stopped  AJRouter           AllJoyn Router Service
Stopped  ALG                Application Layer Gateway Service
Stopped  AppIDSvc           Application Identity
Running  Appinfo            Application Information
Stopped  AppMgmt            Application Management
Stopped  AppReadiness       App Readiness
Stopped  AppVClient         Microsoft App-V Client
Stopped  AppXSvc            AppX Deployment Service (AppXSVC)
Stopped  AssignedAccessM... AssignedAccessManager Service
Running  AudioEndpointBu... Windows Audio Endpoint Builder
Running  Audiosrv           Windows Audio
Stopped  autotimesvc        Cellular Time
Stopped  AxInstSV           ActiveX Installer (AxInstSV)
Stopped  BcastDVRUserSer... GameDVR and Broadcast User Service_...
Stopped  BDESVC             BitLocker Drive Encryption Service
Running  BFE                Base Filtering Engine
Stopped  BITS               Background Intelligent Transfer Ser
```

2nd method (recommended)

Parameters are entered inside the param() always defined first in function

Function show-text {
Param(
$message,
$servicename
)
Write-host "$message" -foregroundcolor red
Get-service -name $servicename
}

To run,

Show-text - message "newparam" -servicename "bits"

```
PS C:\Users\lenovo> Function show-text {
>> Param(
>>            $message,
>>            $servicename
>> )
>> Write-host "$message" -foregroundcolor red
>> Get-service -name $servicename
>> }
>>
PS C:\Users\lenovo> show-text -message "New parameter" -sercvicename "Bits"
New parameter

Status   Name               DisplayName
------   ----               -----------
Stopped  AarSvc_4bf2e02     Agent Activation Runtime_4bf2e02
Running  AdobeARMservice    Adobe Acrobat Update Service
Stopped  AJRouter           AllJoyn Router Service
Stopped  ALG                Application Layer Gateway Service
Stopped  AppIDSvc           Application Identity
Running  Appinfo            Application Information
Stopped  AppMgmt            Application Management
Stopped  AppReadiness       App Readiness
Stopped  AppVClient         Microsoft App-V Client
Running  AppXSvc            AppX Deployment Service (AppXSVC)
Stopped  AssignedAccessM... AssignedAccessManager Service
Running  AudioEndpointBu... Windows Audio Endpoint Builder
Running  Audiosrv           Windows Audio
Stopped  autotimesvc        Cellular Time
Stopped  AxInstSV           ActiveX Installer (AxInstSV)
Stopped  BcastDVRUserSer... GameDVR and Broadcast User Service_...
Stopped  BDESVC             BitLocker Drive Encryption Service
Running  BFE                Base Filtering Engine
Stopped  BITS               Background Intelligent Transfer Ser...
```

Another way

Function show-text {
>> Param(
>> $message = "Hey? How are you??",
>> $servicename
>>)
>> Write-host "$message" -foregroundcolor red
>> Get-service -name $servicename | select-object -last 10
>> }

```
PS C:\Users\lenovo> Function show-text {
>> Param(
>>            $message = "Hey? How are you??",
>>            $servicename
>> )
>> Write-host "$message" -foregroundcolor red
>> Get-service -name $servicename | select-object -last 10
>> }
>>
PS C:\Users\lenovo> show-text
Hey? How are you??

Status   Name               DisplayName
------   ----               -----------
Running  WpnService         Windows Push Notifications System S...
Running  WpnUserService_... Windows Push Notifications User Ser...
Running  wscsvc             Security Center
Running  WSearch            Windows Search
Stopped  wuauserv           Windows Update
Stopped  WwanSvc            WWAN AutoConfig
Stopped  XblAuthManager     Xbox Live Auth Manager
Stopped  XblGameSave        Xbox Live Game Save
Stopped  XboxGipSvc         Xbox Accessory Management Service
Stopped  XboxNetApiSvc      Xbox Live Networking Service
```

Another way

Function show-text {
>> Param(
>> $message = "Hey? How are you??",
>> $servicename
>>)
>> Write-host "$message" -foregroundcolor red
>> Get-service -name $servicename | select-object -last 10
>> }
>>
PS C:\Users\lenovo> **show-text -message "this is custom message"**
this is custom message

```
PS C:\Users\lenovo> Function show-text {
>> Param(
>>            $message = "Hey? How are you??",
>>            $servicename
>> )
>> Write-host "$message" -foregroundcolor red
>> Get-service -name $servicename | select-object -last 10
>> }
>>
PS C:\Users\lenovo> show-text -message "this is custome message"
this is custome message

Status   Name               DisplayName
------   ----               -----------
Running  WpnService         Windows Push Notifications System S...
Running  WpnUserService_... Windows Push Notifications User Ser...
Running  wscsvc             Security Center
Running  WSearch            Windows Search
Stopped  wuauserv           Windows Update
Stopped  WwanSvc            WWAN AutoConfig
Stopped  XblAuthManager     Xbox Live Auth Manager
Stopped  XblGameSave        Xbox Live Game Save
Stopped  XboxGipSvc         Xbox Accessory Management Service
Stopped  XboxNetApiSvc      Xbox Live Networking Service
```

So here we added custom message where we changed the actual value of message and servicename and overwritten with different value.

```
PS C:\Users\lenovo> Function show-text {
>> Param(
>>            $message = "Hey? How are you??",
>>            $servicename = "BITS"
>> )
>> Write-host "$message" -foregroundcolor red
>> Get-service -name $servicename | select-object -last 10
>> }
>>
PS C:\Users\lenovo> show-text -message "this is custome message" $servicename "winrm"
this is custome message

Status   Name               DisplayName
------   ----               -----------
Running  WpnService         Windows Push Notifications System S...
Running  WpnUserService_... Windows Push Notifications User Ser...
Running  wscsvc             Security Center
Running  WSearch            Windows Search
Stopped  wuauserv           Windows Update
Stopped  WwanSvc            WWAN AutoConfig
Stopped  XblAuthManager     Xbox Live Auth Manager
Stopped  XblGameSave        Xbox Live Game Save
Stopped  XboxGipSvc         Xbox Accessory Management Service
Stopped  XboxNetApiSvc      Xbox Live Networking Service
```

Mandatory Parameter [Parameter(mandatory)]

Here in order to make any parameter mandatory we use **[Parameter(mandatory)]**
The value of mandatory parameter need to be entered or should be present or else the function won't run
it effects only for $message parameter as it is only applied to $message, $servicename is not effected.

```
PS C:\Users\lenovo> function show-text {
>> param (
>>   [Parameter(Mandatory)]
>>   $message,
>> $servicename = "bits"
>> )
>> write-host "$message" -foregroundcolor red
>> get-service -name $servicename
>>
>> }
PS C:\Users\lenovo> show-text -message "Mandatory param" -servicename "BITS"
Mandatory param

Status   Name               DisplayName
------   ----               -----------
Stopped  BITS               Background Intelligent Transfer Ser...
```

Variable Scoping in Functions

Global Scopes and Child Scopes
Scopes defined where the parameter can be used in function
$a ="123" - global scope
Function show-demo{
Write-host $a
}

```
PS C:\Users\lenovo> $a = 123
PS C:\Users\lenovo> function show-demo{
>> write-host $a
>> }
PS C:\Users\lenovo> show-demo
123
```

If we try to change the value of $a as 456 inside a show-demo function, the value gets changed inside the function only. If we try to access the value present outside the function we only get it as 123 because it is global scope
Variable Scoping in Functions

```
PS C:\Users\lenovo> function show-demo{
>> $a ="456"
>> write-host $a
>> }
PS C:\Users\lenovo> show-demo
456
PS C:\Users\lenovo> $a
123
```

Function show-demo{
$a =456 #child scope
Write-host $a
}
Here if we access outside the function we get the value of $a as 123 instead of 456